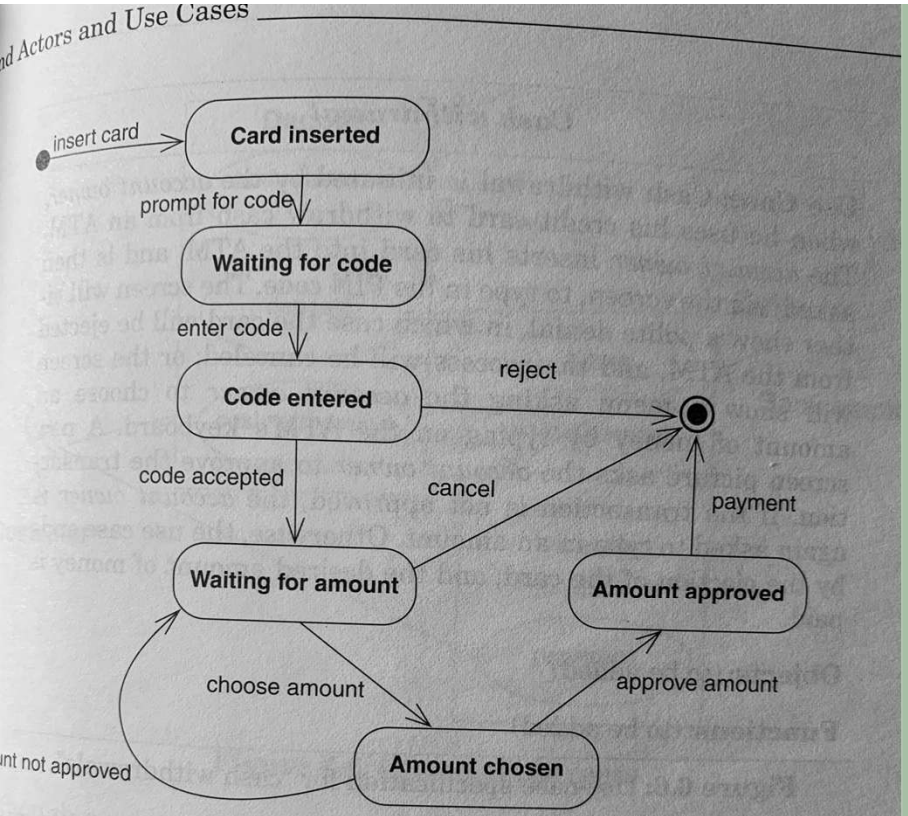


Figure 6.5: Statechart diagram for "cash withdrawal"

target system. We delimit a use case based on the specific act
int and application-domain tasks. The goal is to collect the ma
e ways of using the target system in a few well-chosen use cas
together, your use cases should give an overview; individually th
be abstractions that are both logical and meaningful to the involv

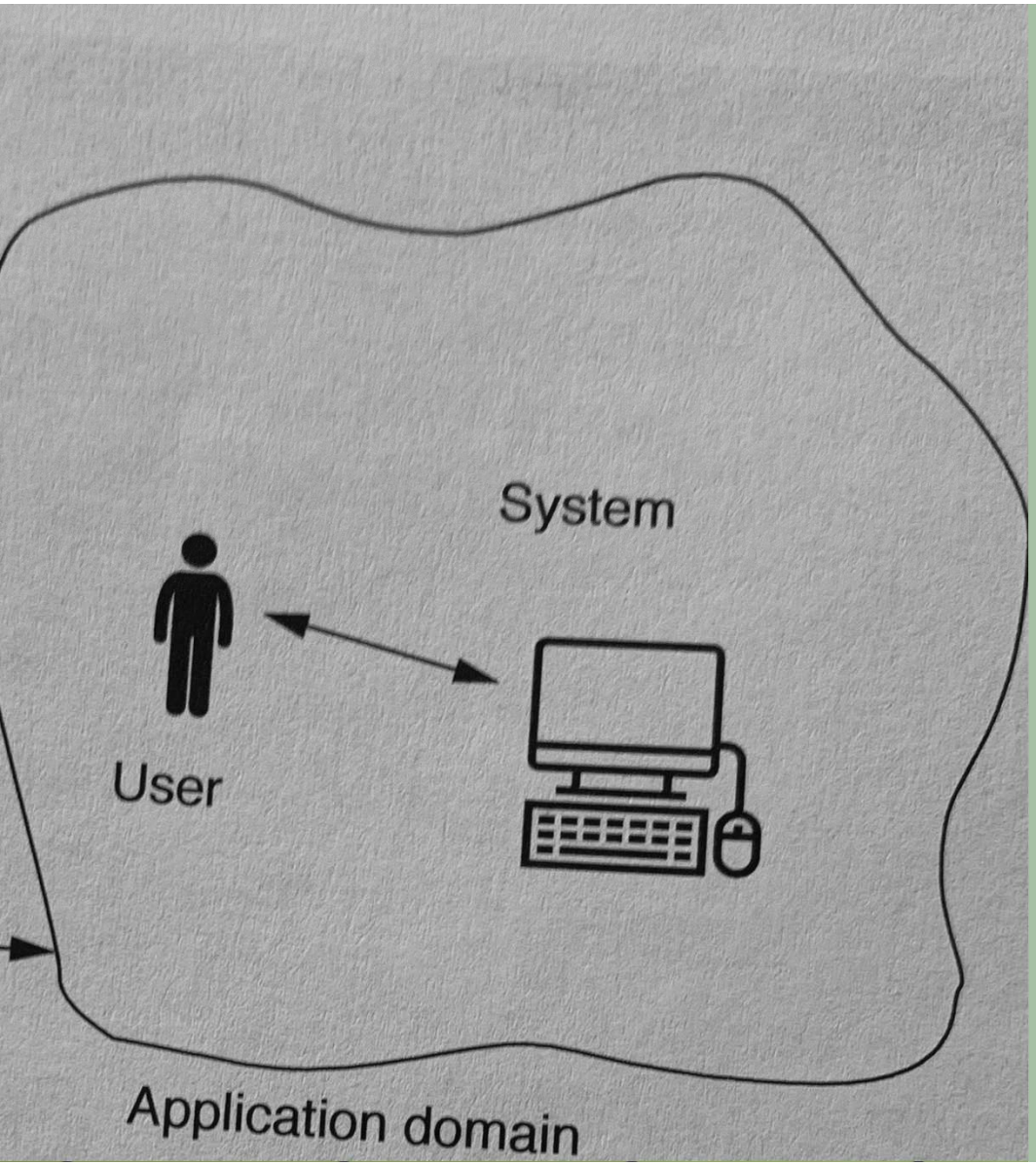
can produce a list of possible use cases by examining the appli
main's tasks. To determine whether or not the use cases are actua

System Development

Session 5 - Application Domain: Usage

Anders Bruun

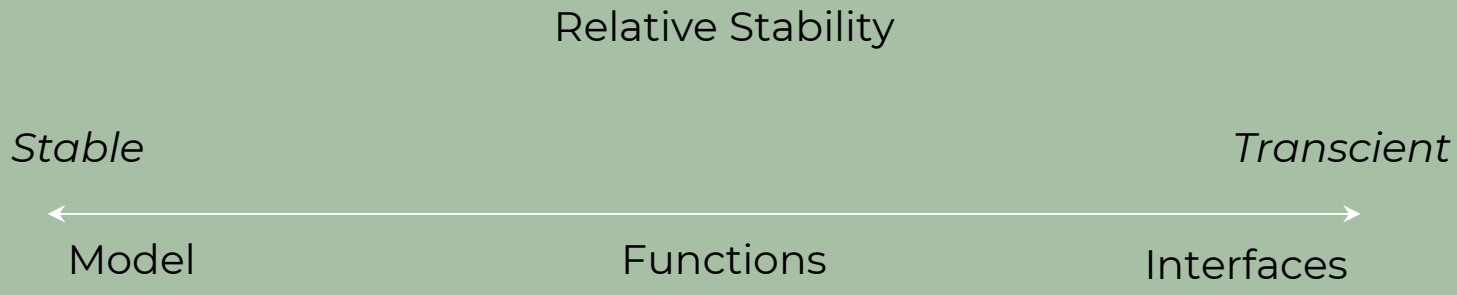
hcc



Application Domain

The organization that administrates, monitors or controls a problem domain

Why Distinguish?



decompose the interface layer into two separate parts: user inte
ystem interface. Figure 10.5 shows this simple and basic architec

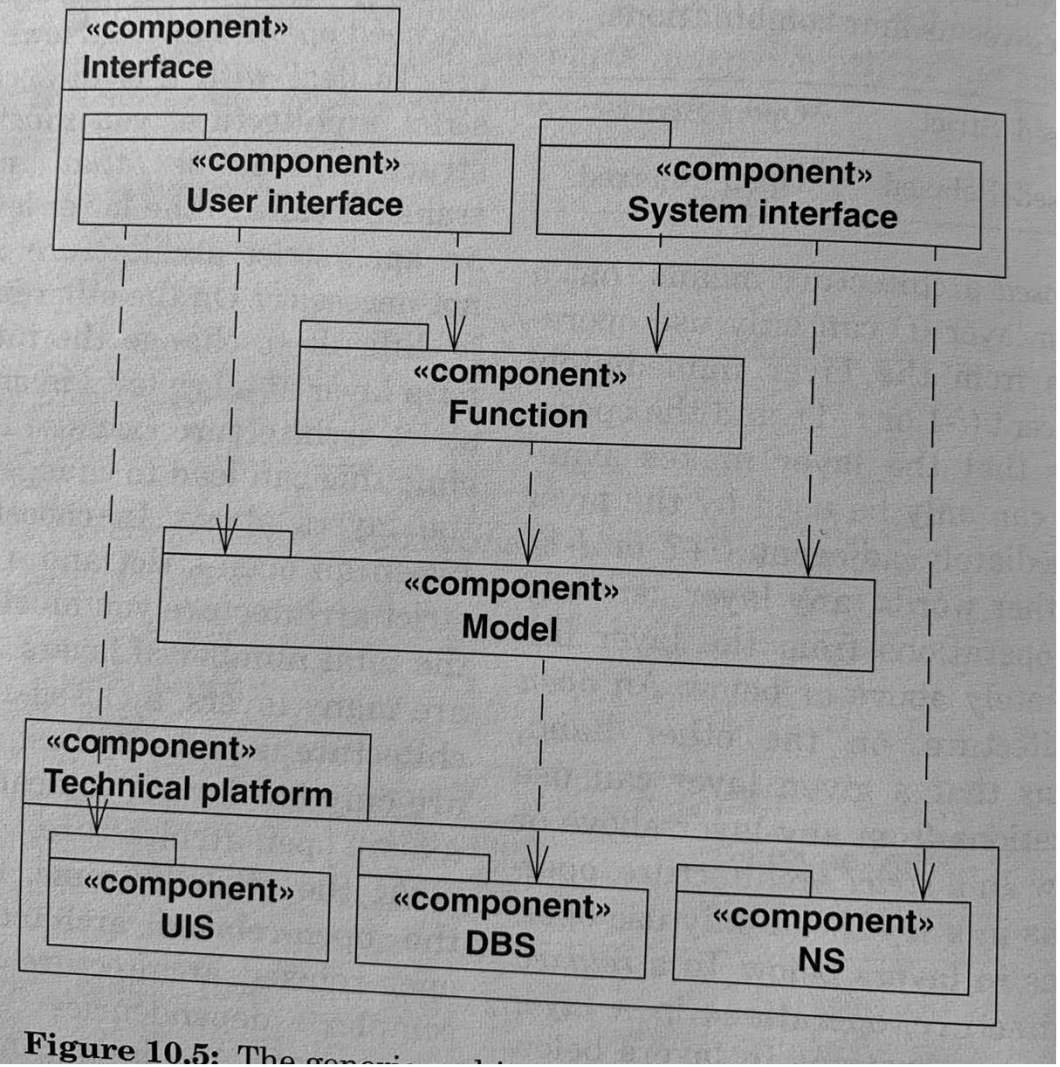


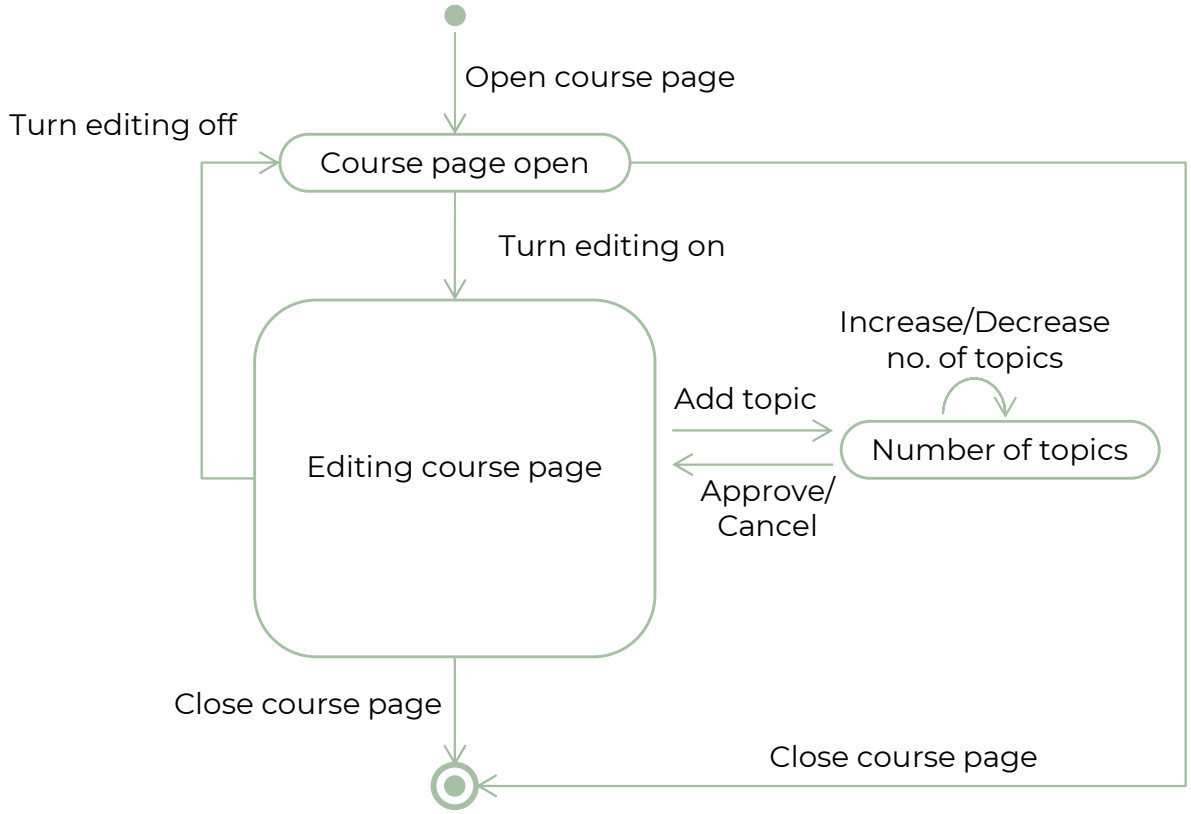
Figure 10.5: The

"When you change the model, you must change the functions and interfaces."

—Mathiassen et al., 2000

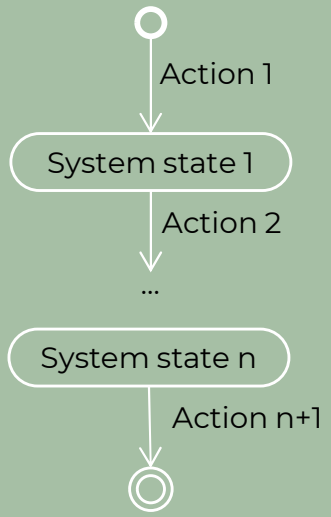
Moodle: Create Session Topic

Use Case: Create session topic is initiated by a *lecturer* when preparing the next lecture. The *lecturer* opens the Moodle page and presses “Turn editing on”. Then selects “Add topics” followed by the number of new topics to be added in the lecture plan. The new topic(s) are then listed ready to be edited.

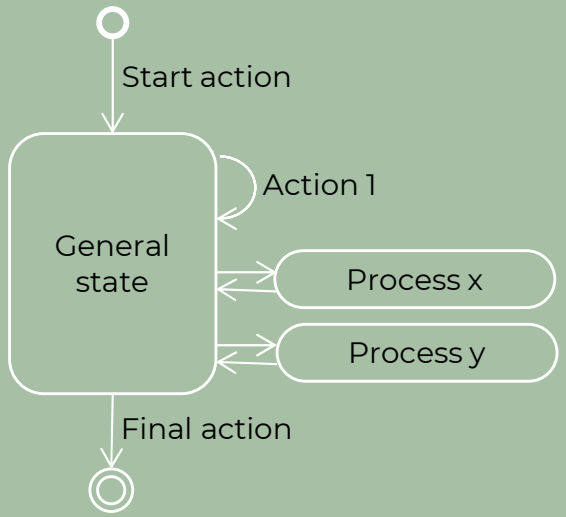


Use Case

A pattern for interaction between the system and actors in the Application Domain



UML State Chart
Procedural Use Case Pattern



UML State Chart
Material Use Case Pattern

Use Case Title

Use case: *Actions* performed by an *actor* that leads to changes in *system states*...

... And so on...

Use Case Specification, brief, precise and structured text focusing on the actor, actions and system states.

Evaluate Systematically

Who and What

Identify individual actors and their system interactions for each use case

Simple and Coherent Whole

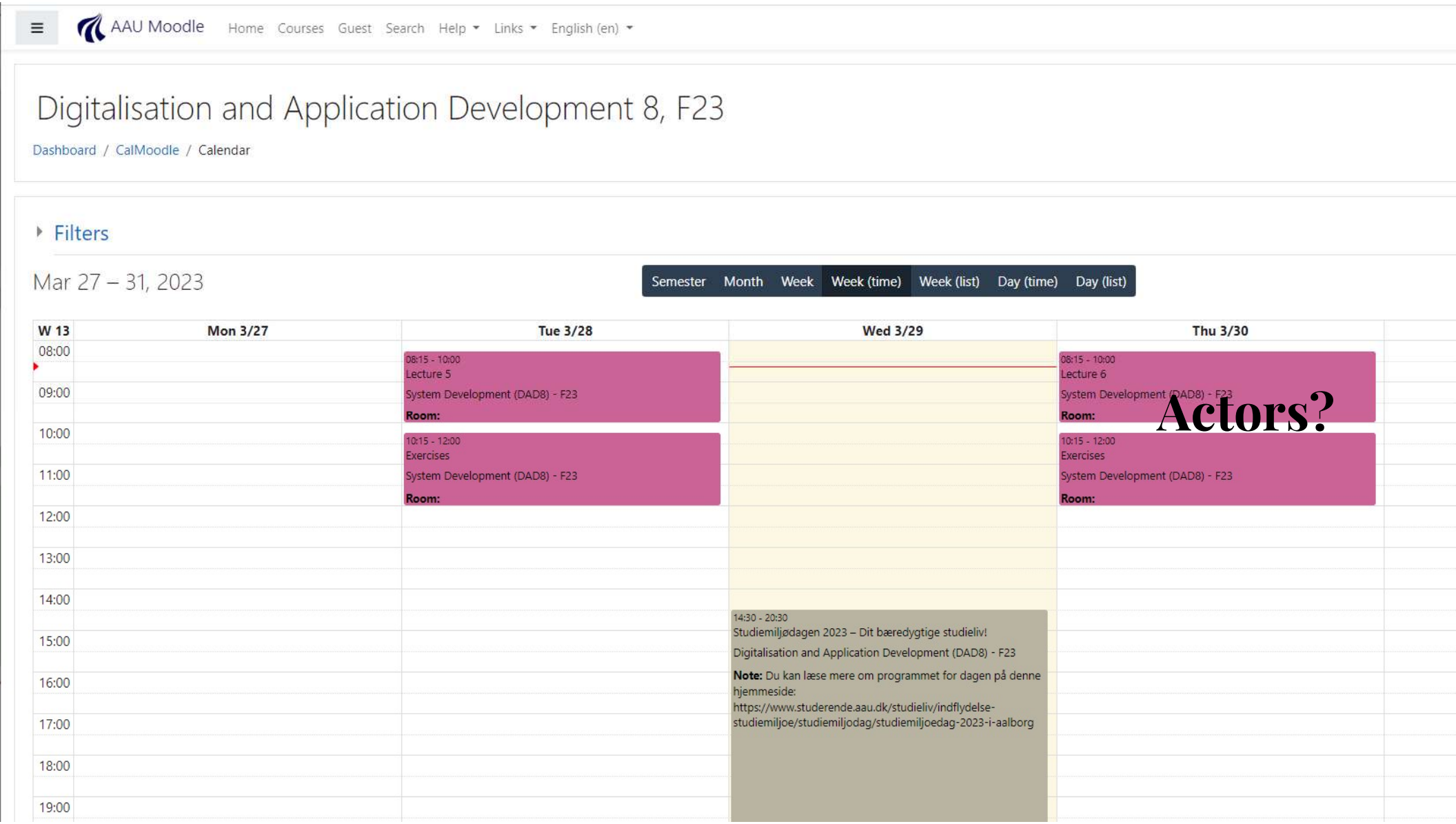
Must be sufficiently detailed, i.e. developers must be able to identify requirements for functions and interfaces

Not too Detailed

Further elaboration happens through detailed user interface designs, e.g. wireframes, prototypes etc.

Actor

An abstraction of users or other systems that interact with the target system


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https://www.studerende.aau.dk/studieliv/indflydelse-studiemiljoe/studiemiljodag/studiemiljoedag-2023-i-aalborg
Actors?

Actor name

Goal: The actor's role in the target system. E.g. "A lecturer needs to manage teaching activities..."

Characteristics: Important aspects of the actor's use of the system. E.g. "There are several lecturers that are supposed to use the system to provide lecture plans with teaching materials in order to align with student schedules and learning goals..."

Examples: General characteristics can be supplemented by concrete examples, e.g. short Persona-like descriptions.

Identify Actors

To identify actors, you must determine the division of labor and the task-related roles in the Application Domain.

Dissimilarities

A criterion for determining different actors is the dissimilarity of roles, as expressed by the use cases in which actors are involved. Much like Classes that share all of their Events would be candidates for merging.

Use Case Diagram

Oerview of relations between actors and use cases.

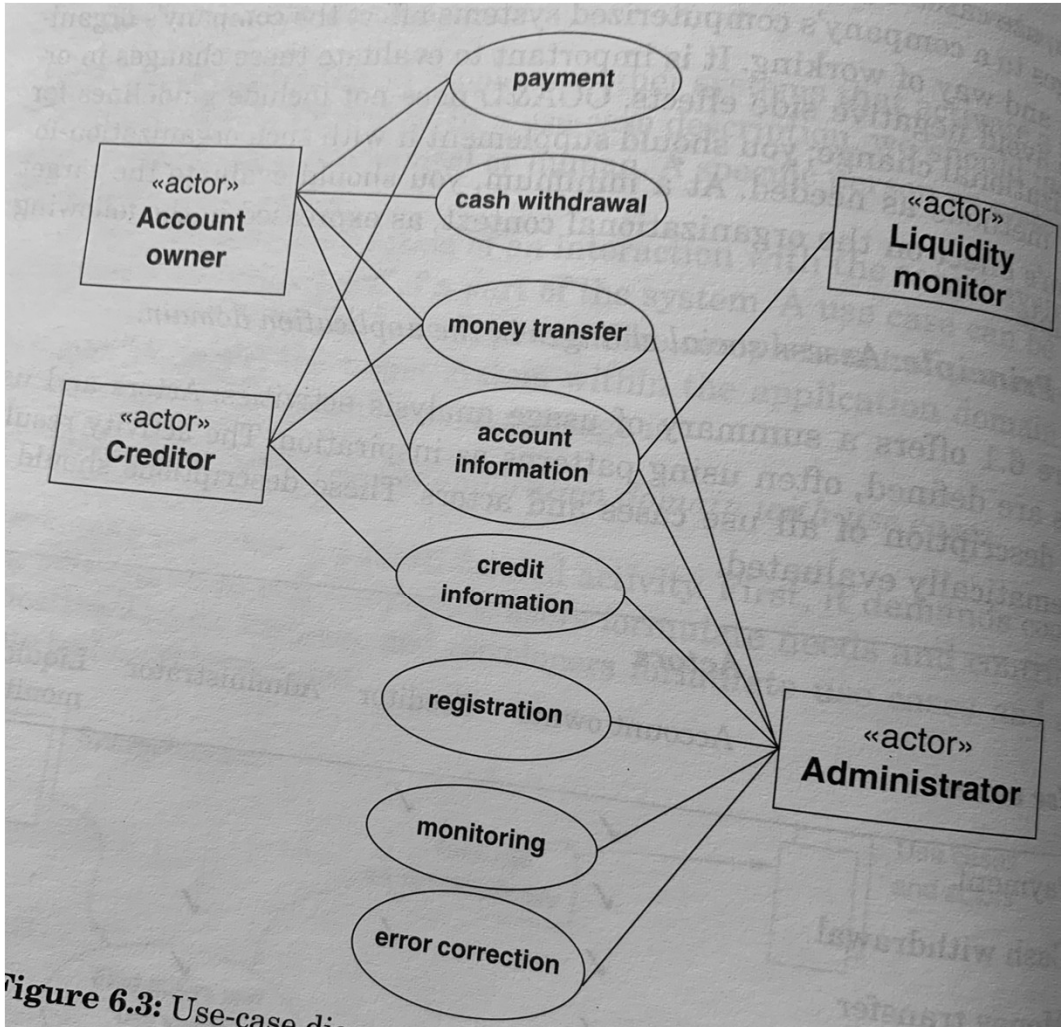


Figure 6.3: Use-case diagram for the automatic payment system

actor table or a use-case diagram. We prefer th
less space; UML recommends th
count owner is

Use Cases in the Application Domain

Exercise 5.1

Define what you find to be the central Use Case for the F-klubben case. Create a UML state-transition diagram to describe the interaction between an actor and the system for this Use Case. Think in terms of defining the various states that the system can be in during the interaction.

Note that this is the exact same type of UML diagram used to define the Behavioural Patterns of Classes within the Problem Domain, yet, now we use this type of diagram in the Application Domain analysis to specify interactions between the system and the actors.

The Use Case Diagram – An Overview Relations between Actors and Use Cases

Exercise 5.2

List 3 potential Use Cases of the F-Klubben case (for this exercise you do not need to specify all 3 Use Cases through UML State Chart Diagrams).

Identify and describe at least 3 Actors that are interacting with the F-Klub system and create a Use Case Diagram to illustrate the relation between use cases and actors.

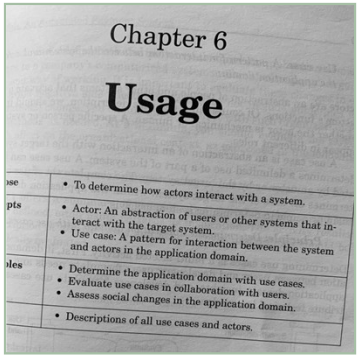
Use Cases and Use Case Diagram for your Projects

Exercise 5.3

Identify and describe Actors and Use Cases for your semester projects. Make sure to evaluate Use Cases systematically (see slide 7) and identify and describe the Actors in sufficient detail (slide 14).

Make a Use Case Diagram to provide an overview of the relation between Actors and Use Cases.

Until we meet again


Chapter 6

Usage

ase	• To determine how actors interact with a system.
pts	• Actor: An abstraction of users or other systems that interact with the target system. • Use case: A pattern for interaction between the system and actors in the application domain.
ples	• Determine the application domain with use cases. • Evaluate use cases in collaboration with users. • Assess social changes in the application domain.
	• Descriptions of all use cases and actors.

